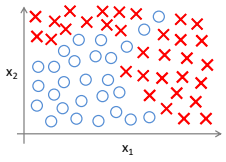
Applied Machine Learning

Neural Networks: Representation

Non-linear hypotheses

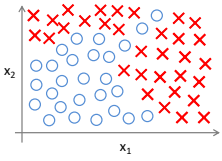
1

Non-linear Classification



2

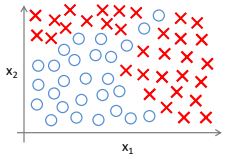
Non-linear Classification



Draw a non-linear boundary?

3

Non-linear Classification



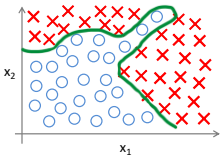
Draw a non-linear boundary?

Apply Logistic Regression with Polynomial terms

$$g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1 x_2 + \theta_4 x_1^2 x_2 + \theta_5 x_1^3 x_2 + \theta_6 x_1 x_2^2 + \dots)$$

4

Non-linear Classification



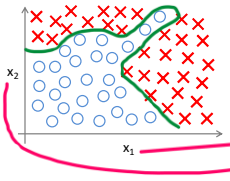
Draw a non-linear boundary?

Apply Logistic Regression with Polynomial terms

$$g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1 x_2 + \theta_4 x_1^2 x_2 + \theta_5 x_1^3 x_2 + \theta_6 x_1 x_2^2 + \dots)$$

5

Non-linear Classification



Draw a non-linear boundary?

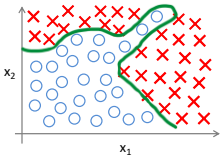
Apply Logistic Regression with Polynomial terms

$$g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1 x_2 + \theta_4 x_1^2 x_2 + \theta_5 x_1^3 x_2 + \theta_6 x_1 x_2^2 + \dots)$$

Works well for a small number features

6

Non-linear Classification



Draw a non-linear boundary?

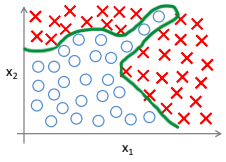
Apply Logistic Regression with Polynomial terms

$$g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1 x_2 + \theta_4 x_1^2 x_2 + \theta_5 x_1^3 x_2 + \theta_6 x_1 x_2^2 + \dots)$$

x_1 = size
x_2 = # bedrooms
x_3 = # floors
x_4 = age
...
x_{100}

7

Non-linear Classification



Draw a non-linear boundary?

Apply Logistic Regression with Polynomial terms

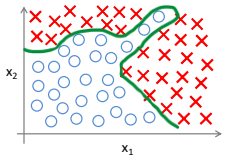
$$g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1 x_2 + \theta_4 x_1^2 x_2 + \theta_5 x_1^3 x_2 + \theta_6 x_1 x_2^2 + \dots)$$

x_1 = size
x_2 = # bedrooms
x_3 = # floors
x_4 = age
...
x_{100}

$x = 100$

8

Non-linear Classification



x_1 = size
x_2 = # bedrooms
x_3 = # floors
x_4 = age
...
x_{100}

x = 100

Draw a non-linear boundary?

Apply Logistic Regression with Polynomial terms

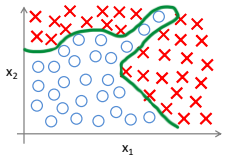
$$g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1 x_2 + \theta_4 x_1^2 x_2 + \theta_5 x_1^3 x_2 + \theta_6 x_1 x_2^2 + \dots)$$

$x_1^2, x_1x_2, x_1x_3, x_1x_4, \dots, x_1x_{100}$

$x_2^2, x_2x_3, x_1x_4, \dots, x_1x_{100}$

9

Non-linear Classification



x_1 = size
x_2 = # bedrooms
x_3 = # floors
x_4 = age
...
x_{100}

x = 100

Draw a non-linear boundary?

Apply Logistic Regression with Polynomial terms

$$g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1 x_2 + \theta_4 x_1^2 x_2 + \theta_5 x_1^3 x_2 + \theta_6 x_1 x_2^2 + \dots)$$

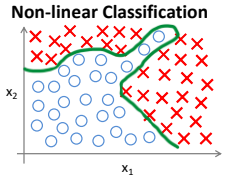
$x_1^2, x_1x_2, x_1x_3, x_1x_4, \dots, x_1x_{100}$

$x_2^2, x_2x_3, x_1x_4, \dots, x_1x_{100}$

≈ 5000 *features*

10

Non-linear Classification



x_1 = size
x_2 = # bedrooms
x_3 = # floors
x_4 = age
...
x_{100}

x = 100

Draw a non-linear boundary?

Apply Logistic Regression with Polynomial terms

$$g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1 x_2 + \theta_4 x_1^2 x_2 + \theta_5 x_1^3 x_2 + \theta_6 x_1 x_2^2 + \dots)$$

$x_1^2, x_1x_2, x_1x_3, x_1x_4, \dots, x_1x_{100}$

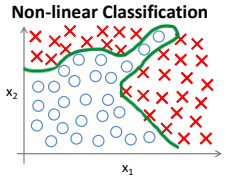
$x_2^2, x_2x_3, x_1x_4, \dots, x_1x_{100}$

≈ 5000 *features*

Do you see a problem???

11

Non-linear Classification



x_1 = size
x_2 = # bedrooms
x_3 = # floors
x_4 = age
...
x_{100}

x = 100

Draw a non-linear boundary?

Apply Logistic Regression with Polynomial terms

$$g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1 x_2 + \theta_4 x_1^2 x_2 + \theta_5 x_1^3 x_2 + \theta_6 x_1 x_2^2 + \dots)$$

$x_1^2, x_1x_2, x_1x_3, x_1x_4, \dots, x_1x_{100}$

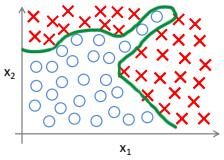
$x_2^2, x_2x_3, x_1x_4, \dots, x_1x_{100}$

≈ 5000 *features*

Do you see a problem???

12

Non-linear Classification



x_1 = size
x_2 = # bedrooms
x_3 = # floors
x_4 = age
$\cdots$
x_{100}

$x = 100$

Draw a non-linear boundary?

Apply Logistic Regression with Polynomial terms

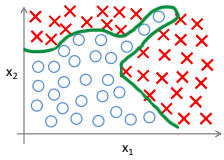
$$g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1 x_2 + \theta_4 x_1^2 x_2 + \theta_5 x_1^3 x_2 + \theta_6 x_1 x_2^2 + \dots)$$

$x_1^2, x_1x_2, x_1x_3, x_1x_4, \dots, x_1x_{100}$
$x_2^2, x_2x_3, x_1x_4, \dots, x_1x_{100}$
$\approx$ **5000 *features***

$x_1^2, x_2^2, x_3^2, \dots, x_{100}^2$
Only quadratic features, but the boundary will be ellipse

13

Non-linear Classification



x_1 = size
x_2 = # bedrooms
x_3 = # floors
x_4 = age
$\cdots$
x_{100}

$x = 100$

Draw a non-linear boundary?

Apply Logistic Regression with Polynomial terms

$$g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1 x_2 + \theta_4 x_1^2 x_2 + \theta_5 x_1^3 x_2 + \theta_6 x_1 x_2^2 + \dots)$$

$x_1^2, x_1x_2, x_1x_3, x_1x_4, \dots, x_1x_{100}$
$x_2^2, x_2x_3, x_1x_4, \dots, x_1x_{100}$
$\approx$ **5000 *features***

$x_1^2, x_2^2, x_3^2, \dots, x_{100}^2$
Only quadratic features, but the boundary will be ellipse

$x_1x_2x_3, x_1^2x_2, x_1x_2x_{17}, \dots \approx$ **170,000 *features***

14

What is this?

You see this:



15

What is this?

You see this:



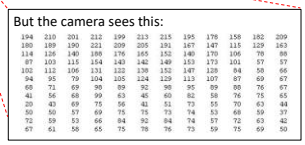
16

What is this?

You see this:



But the camera sees this:



17

Computer Vision: Car detection

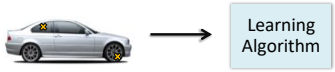


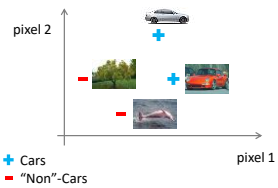
Testing:



What is this?

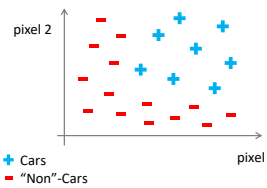
18





19





50 x 50 pixel images→ 2500 pixels

$n = 2500$ (7500 if RGB)

$$x = \begin{bmatrix} \text{pixel 1 intensity} \\ \text{pixel 2 intensity} \\ \vdots \\ \text{pixel 2500 intensity} \end{bmatrix}$$

Quadratic features ($x_i \times x_j$): ≈3 million features

20

Applied Machine Learning

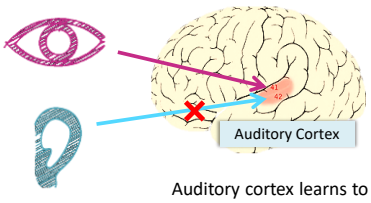
Neural Networks: Representation

Neurons and the brain

Neural Networks

Origins: Algorithms that try to mimic the brain.
Was very widely used in 80s and early 90s; popularity diminished in late 90s.
Recent resurgence: State-of-the-art technique for many applications

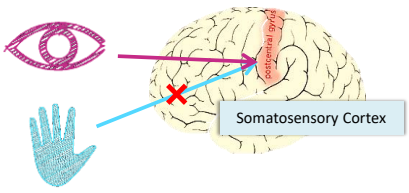
The "one learning algorithm" hypothesis



Auditory cortex learns to see

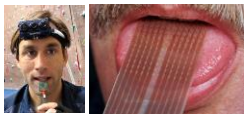
[Roe et al., 1992]

The "one learning algorithm" hypothesis



[Metin & Frost, 1989]

Sensor representations in the brain



Seeing with your tongue



Human echolocation (sonar)



Haptic belt: Direction sense



Implanting a 3rd eye

[BrainPort; Welsh & Blasch, 1997; Nagel et al., 2005; Constantine-Paton & Law, 2009]

25

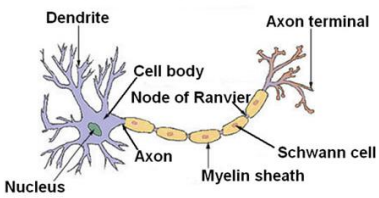
Applied Machine Learning

Neural Networks: Representation

Model representation I

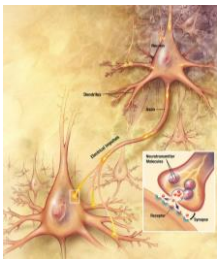
26

Neuron in the brain



27

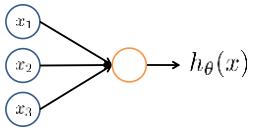
Neurons in the brain



[Credit: US National Institutes of Health, National Institute on Aging]

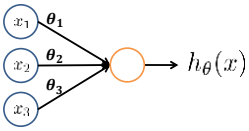
28

Neuron model: Logistic unit



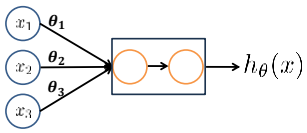
29

Neuron model: Logistic unit



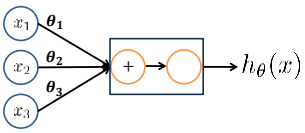
30

Neuron model: Logistic unit



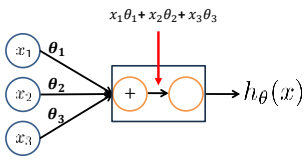
31

Neuron model: Logistic unit



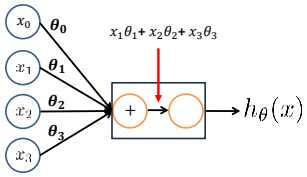
32

Neuron model: Logistic unit



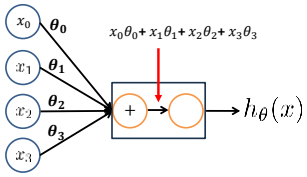
33

Neuron model: Logistic unit



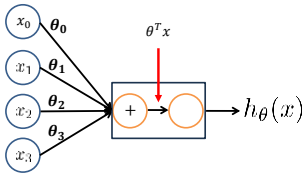
34

Neuron model: Logistic unit



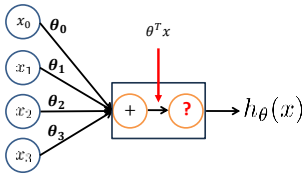
35

Neuron model: Logistic unit



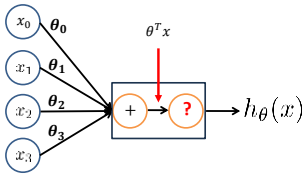
36

Neuron model: Logistic unit



37

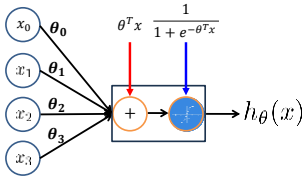
Neuron model: Logistic unit



Sigmoid (logistic) activation function.

38

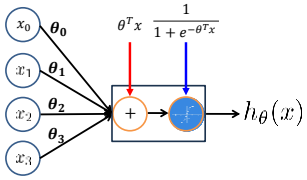
Neuron model: Logistic unit

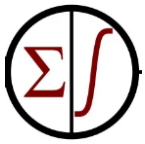


Sigmoid (logistic) activation function.

39

Neuron model: Logistic unit

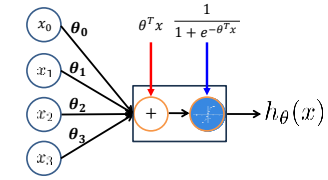


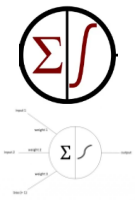


Sigmoid (logistic) activation function.

40

Neuron model: Logistic unit

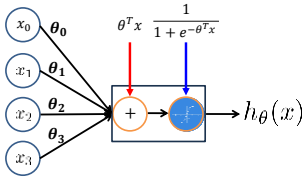


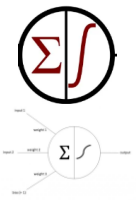


Sigmoid (logistic) activation function.

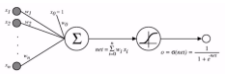
41

Neuron model: Logistic unit



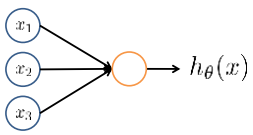


Sigmoid (logistic) activation function.



42

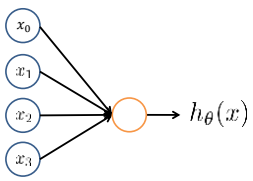
Neuron model: Logistic unit



Sigmoid (logistic) activation function.

43

Neuron model: Logistic unit

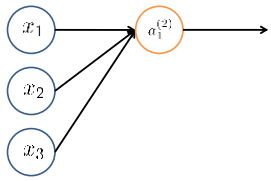


$$x = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad \theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix}$$

Sigmoid (logistic) activation function.

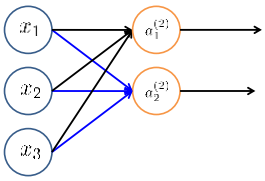
44

Neural Network



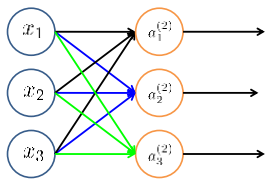
45

Neural Network



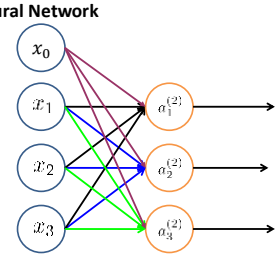
46

Neural Network



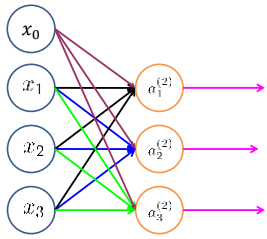
47

Neural Network



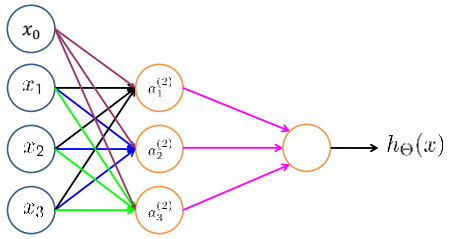
48

Neural Network



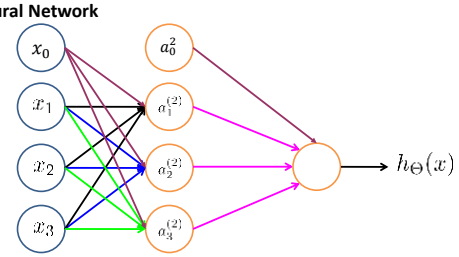
49

Neural Network



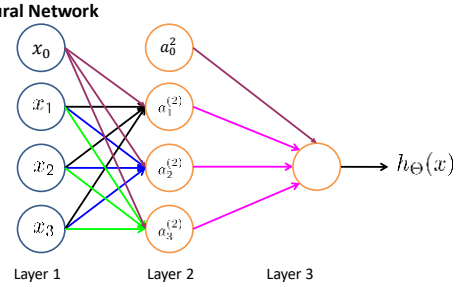
50

Neural Network

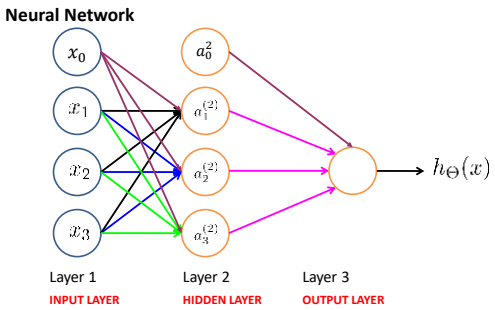


51

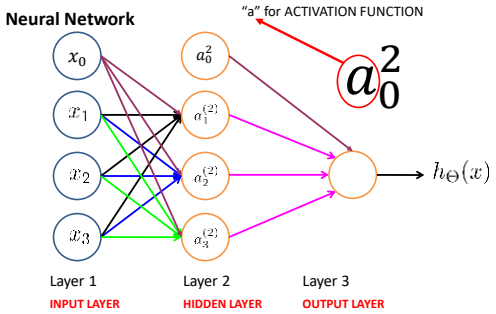
Neural Network



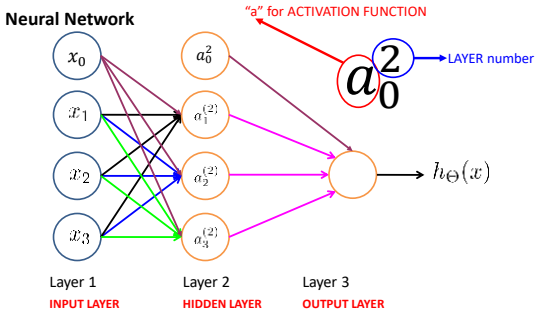
52


Neural Network
x_0
x_1
x_2
x_3
a_0^2
$a_1^{(2)}$
$a_2^{(2)}$
$a_3^{(2)}$
$h_\Theta(x)$
Layer 1
INPUT LAYER
Layer 2
HIDDEN LAYER
Layer 3
OUTPUT LAYER

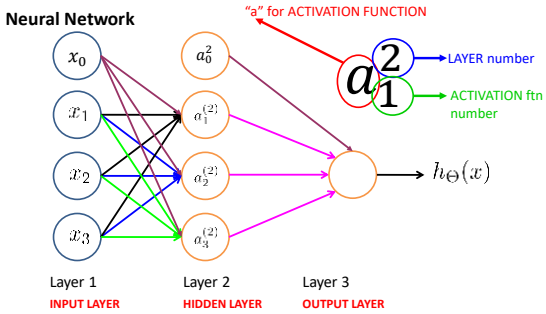
53


Neural Network
"a" for ACTIVATION FUNCTION
a_0^2
x_0
x_1
x_2
x_3
a_0^2
$a_1^{(2)}$
$a_2^{(2)}$
$a_3^{(2)}$
$h_\Theta(x)$
Layer 1
INPUT LAYER
Layer 2
HIDDEN LAYER
Layer 3
OUTPUT LAYER

54

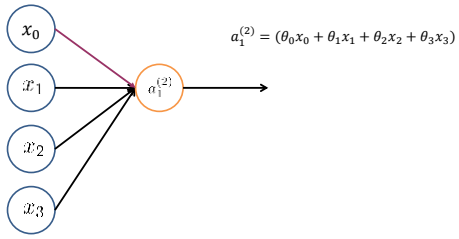

Neural Network
"a" for ACTIVATION FUNCTION
a_0^2
LAYER number
x_0
x_1
x_2
x_3
a_0^2
$a_1^{(2)}$
$a_2^{(2)}$
$a_3^{(2)}$
$h_\Theta(x)$
Layer 1
INPUT LAYER
Layer 2
HIDDEN LAYER
Layer 3
OUTPUT LAYER

55


Neural Network
"a" for ACTIVATION FUNCTION
a_1^2
LAYER number
ACTIVATION ftn number
x_0
x_1
x_2
x_3
a_0^2
$a_1^{(2)}$
$a_2^{(2)}$
$a_3^{(2)}$
$h_\Theta(x)$
Layer 1
INPUT LAYER
Layer 2
HIDDEN LAYER
Layer 3
OUTPUT LAYER

56

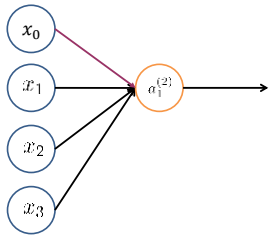
Neural Network



57

Neural Network

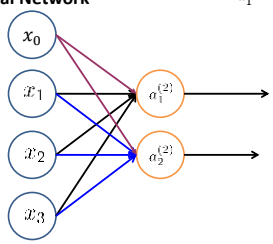
$a_1^{(2)} = g(\theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3)$



58

Neural Network

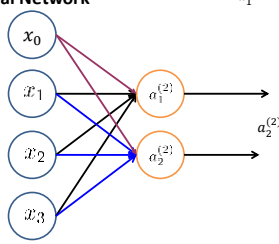
$a_1^{(2)} = g(\theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3)$



59

Neural Network

$a_1^{(2)} = g(\theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3)$

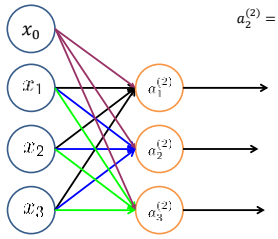


60

Neural Network

$a_1^{(2)} = g(\theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3)$

$a_2^{(2)} = g(\theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3)$



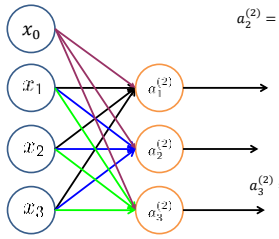
61

Neural Network

$a_1^{(2)} = g(\theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3)$

$a_2^{(2)} = g(\theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3)$

$a_3^{(2)} = g(\theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3)$



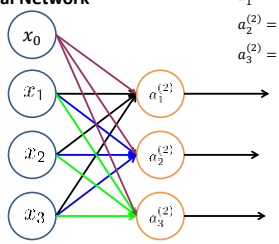
62

Neural Network

$a_1^{(2)} = g(\theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3)$

$a_2^{(2)} = g(\theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3)$

$a_3^{(2)} = g(\theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3)$



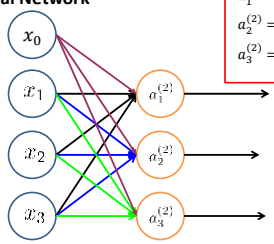
63

Neural Network

$a_1^{(2)} = g(\theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3)$

$a_2^{(2)} = g(\theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3)$

$a_3^{(2)} = g(\theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3)$

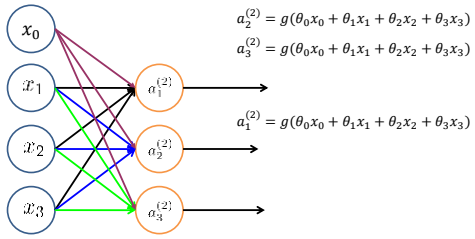


LOOK SIMILAR???



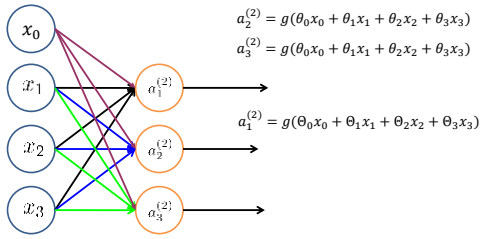
64

Neural Network



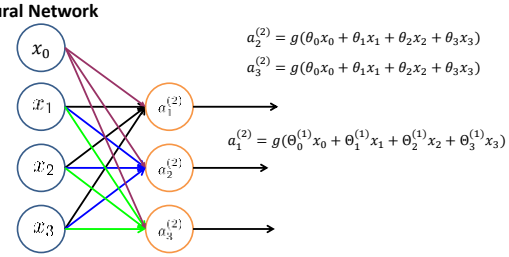
65

Neural Network



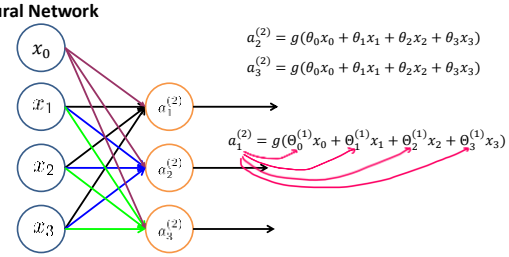
66

Neural Network



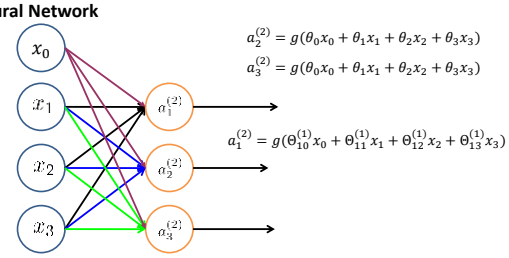
67

Neural Network



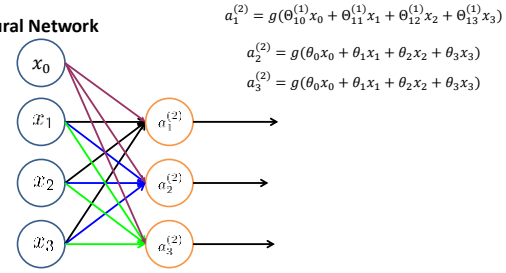
68

Neural Network



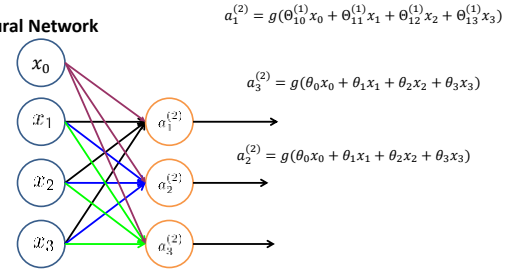
69

Neural Network



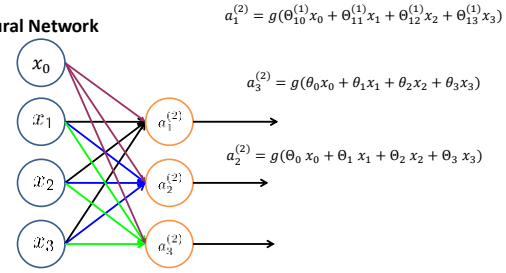
70

Neural Network

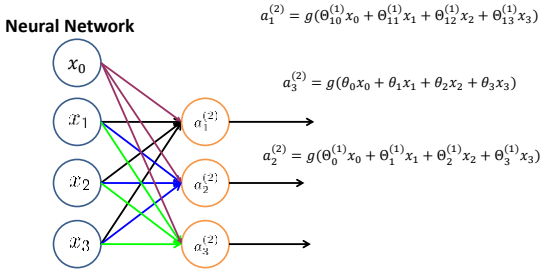


71

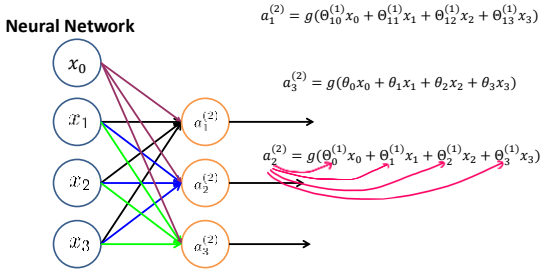
Neural Network



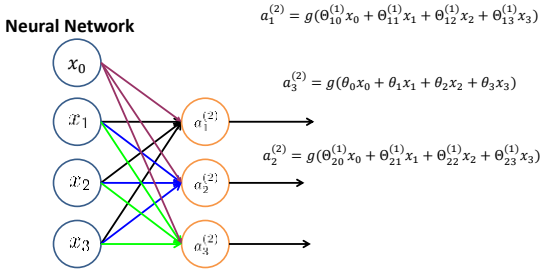
72


Neural Network
$a_1^{(2)} = g(\Theta_{10}^{(1)}x_0 + \Theta_{11}^{(1)}x_1 + \Theta_{12}^{(1)}x_2 + \Theta_{13}^{(1)}x_3)$
$a_3^{(2)} = g(\theta_0x_0 + \theta_1x_1 + \theta_2x_2 + \theta_3x_3)$
$a_2^{(2)} = g(\Theta_0^{(1)}x_0 + \Theta_1^{(1)}x_1 + \Theta_2^{(1)}x_2 + \Theta_3^{(1)}x_3)$
x_0
x_1
x_2
x_3
$a_1^{(2)}$
$a_2^{(2)}$
$a_3^{(2)}$

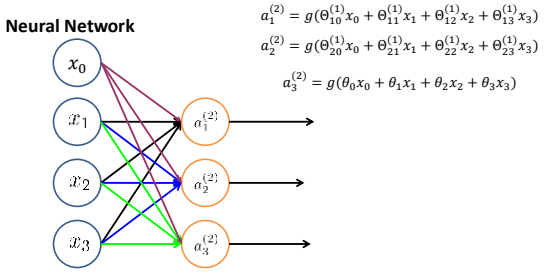
73


Neural Network
$a_1^{(2)} = g(\Theta_{10}^{(1)}x_0 + \Theta_{11}^{(1)}x_1 + \Theta_{12}^{(1)}x_2 + \Theta_{13}^{(1)}x_3)$
$a_3^{(2)} = g(\theta_0x_0 + \theta_1x_1 + \theta_2x_2 + \theta_3x_3)$
$a_2^{(2)} = g(\Theta_0^{(1)}x_0 + \Theta_1^{(1)}x_1 + \Theta_2^{(1)}x_2 + \Theta_3^{(1)}x_3)$
x_0
x_1
x_2
x_3
$a_1^{(2)}$
$a_2^{(2)}$
$a_3^{(2)}$

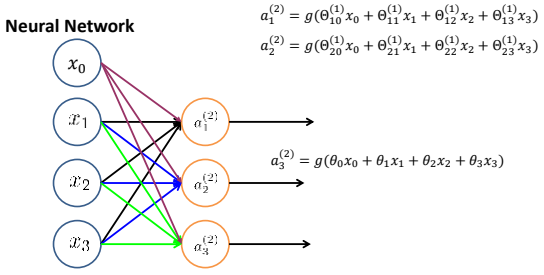
74


Neural Network
$a_1^{(2)} = g(\Theta_{10}^{(1)}x_0 + \Theta_{11}^{(1)}x_1 + \Theta_{12}^{(1)}x_2 + \Theta_{13}^{(1)}x_3)$
$a_3^{(2)} = g(\theta_0x_0 + \theta_1x_1 + \theta_2x_2 + \theta_3x_3)$
$a_2^{(2)} = g(\Theta_{20}^{(1)}x_0 + \Theta_{21}^{(1)}x_1 + \Theta_{22}^{(1)}x_2 + \Theta_{23}^{(1)}x_3)$
x_0
x_1
x_2
x_3
$a_1^{(2)}$
$a_2^{(2)}$
$a_3^{(2)}$

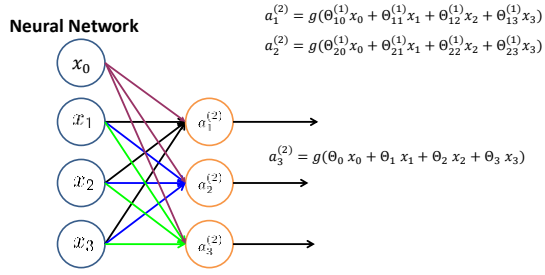
75


Neural Network
$a_1^{(2)} = g(\Theta_{10}^{(1)}x_0 + \Theta_{11}^{(1)}x_1 + \Theta_{12}^{(1)}x_2 + \Theta_{13}^{(1)}x_3)$
$a_2^{(2)} = g(\Theta_{20}^{(1)}x_0 + \Theta_{21}^{(1)}x_1 + \Theta_{22}^{(1)}x_2 + \Theta_{23}^{(1)}x_3)$
$a_3^{(2)} = g(\theta_0x_0 + \theta_1x_1 + \theta_2x_2 + \theta_3x_3)$
x_0
x_1
x_2
x_3
$a_1^{(2)}$
$a_2^{(2)}$
$a_3^{(2)}$

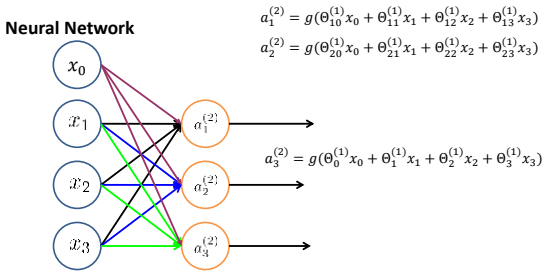
76


Neural Network
$a_1^{(2)} = g(\Theta_{10}^{(1)}x_0 + \Theta_{11}^{(1)}x_1 + \Theta_{12}^{(1)}x_2 + \Theta_{13}^{(1)}x_3)$
$a_2^{(2)} = g(\Theta_{20}^{(1)}x_0 + \Theta_{21}^{(1)}x_1 + \Theta_{22}^{(1)}x_2 + \Theta_{23}^{(1)}x_3)$
$a_3^{(2)} = g(\theta_0x_0 + \theta_1x_1 + \theta_2x_2 + \theta_3x_3)$
x_0
x_1
x_2
x_3
$a_1^{(2)}$
$a_2^{(2)}$
$a_3^{(2)}$

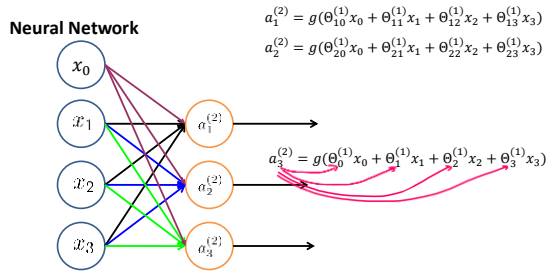
77


Neural Network
$a_1^{(2)} = g(\Theta_{10}^{(1)}x_0 + \Theta_{11}^{(1)}x_1 + \Theta_{12}^{(1)}x_2 + \Theta_{13}^{(1)}x_3)$
$a_2^{(2)} = g(\Theta_{20}^{(1)}x_0 + \Theta_{21}^{(1)}x_1 + \Theta_{22}^{(1)}x_2 + \Theta_{23}^{(1)}x_3)$
$a_3^{(2)} = g(\Theta_0 x_0 + \Theta_1 x_1 + \Theta_2 x_2 + \Theta_3 x_3)$
x_0
x_1
x_2
x_3
$a_1^{(2)}$
$a_2^{(2)}$
$a_3^{(2)}$

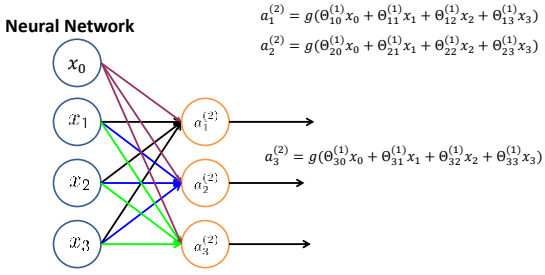
78


Neural Network
$a_1^{(2)} = g(\Theta_{10}^{(1)}x_0 + \Theta_{11}^{(1)}x_1 + \Theta_{12}^{(1)}x_2 + \Theta_{13}^{(1)}x_3)$
$a_2^{(2)} = g(\Theta_{20}^{(1)}x_0 + \Theta_{21}^{(1)}x_1 + \Theta_{22}^{(1)}x_2 + \Theta_{23}^{(1)}x_3)$
$a_3^{(2)} = g(\Theta_0^{(1)}x_0 + \Theta_1^{(1)}x_1 + \Theta_2^{(1)}x_2 + \Theta_3^{(1)}x_3)$
x_0
x_1
x_2
x_3
$a_1^{(2)}$
$a_2^{(2)}$
$a_3^{(2)}$

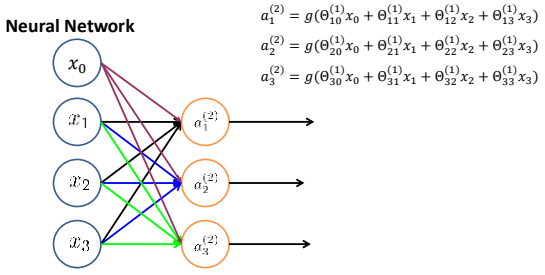
79


Neural Network
$a_1^{(2)} = g(\Theta_{10}^{(1)}x_0 + \Theta_{11}^{(1)}x_1 + \Theta_{12}^{(1)}x_2 + \Theta_{13}^{(1)}x_3)$
$a_2^{(2)} = g(\Theta_{20}^{(1)}x_0 + \Theta_{21}^{(1)}x_1 + \Theta_{22}^{(1)}x_2 + \Theta_{23}^{(1)}x_3)$
$a_3^{(2)} = g(\Theta_0^{(1)}x_0 + \Theta_1^{(1)}x_1 + \Theta_2^{(1)}x_2 + \Theta_3^{(1)}x_3)$
x_0
x_1
x_2
x_3
$a_1^{(2)}$
$a_2^{(2)}$
$a_3^{(2)}$

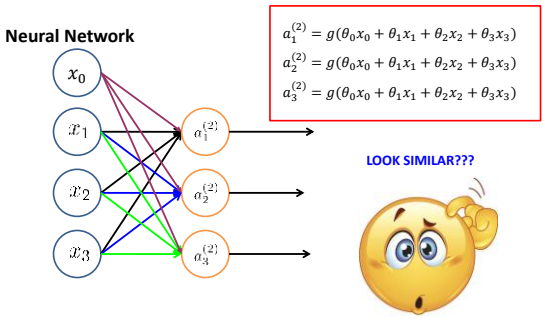
80


Neural Network
$a_1^{(2)} = g(\Theta_{10}^{(1)}x_0 + \Theta_{11}^{(1)}x_1 + \Theta_{12}^{(1)}x_2 + \Theta_{13}^{(1)}x_3)$
$a_2^{(2)} = g(\Theta_{20}^{(1)}x_0 + \Theta_{21}^{(1)}x_1 + \Theta_{22}^{(1)}x_2 + \Theta_{23}^{(1)}x_3)$
$a_3^{(2)} = g(\Theta_{30}^{(1)}x_0 + \Theta_{31}^{(1)}x_1 + \Theta_{32}^{(1)}x_2 + \Theta_{33}^{(1)}x_3)$

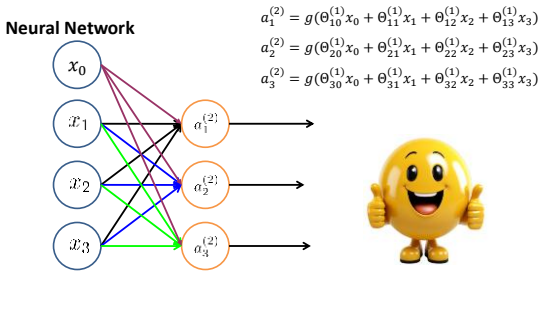
81


Neural Network
$a_1^{(2)} = g(\Theta_{10}^{(1)}x_0 + \Theta_{11}^{(1)}x_1 + \Theta_{12}^{(1)}x_2 + \Theta_{13}^{(1)}x_3)$
$a_2^{(2)} = g(\Theta_{20}^{(1)}x_0 + \Theta_{21}^{(1)}x_1 + \Theta_{22}^{(1)}x_2 + \Theta_{23}^{(1)}x_3)$
$a_3^{(2)} = g(\Theta_{30}^{(1)}x_0 + \Theta_{31}^{(1)}x_1 + \Theta_{32}^{(1)}x_2 + \Theta_{33}^{(1)}x_3)$

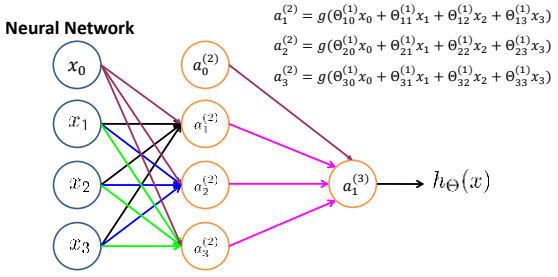
82


Neural Network
$a_1^{(2)} = g(\theta_0x_0 + \theta_1x_1 + \theta_2x_2 + \theta_3x_3)$
$a_2^{(2)} = g(\theta_0x_0 + \theta_1x_1 + \theta_2x_2 + \theta_3x_3)$
$a_3^{(2)} = g(\theta_0x_0 + \theta_1x_1 + \theta_2x_2 + \theta_3x_3)$
LOOK SIMILAR???

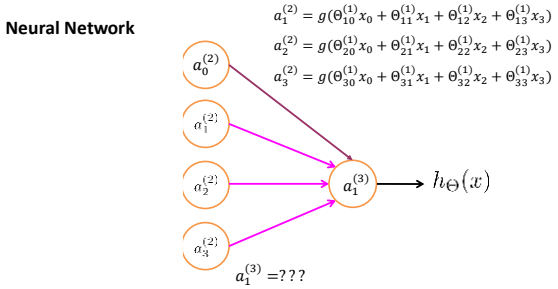
83


Neural Network
$a_1^{(2)} = g(\Theta_{10}^{(1)}x_0 + \Theta_{11}^{(1)}x_1 + \Theta_{12}^{(1)}x_2 + \Theta_{13}^{(1)}x_3)$
$a_2^{(2)} = g(\Theta_{20}^{(1)}x_0 + \Theta_{21}^{(1)}x_1 + \Theta_{22}^{(1)}x_2 + \Theta_{23}^{(1)}x_3)$
$a_3^{(2)} = g(\Theta_{30}^{(1)}x_0 + \Theta_{31}^{(1)}x_1 + \Theta_{32}^{(1)}x_2 + \Theta_{33}^{(1)}x_3)$

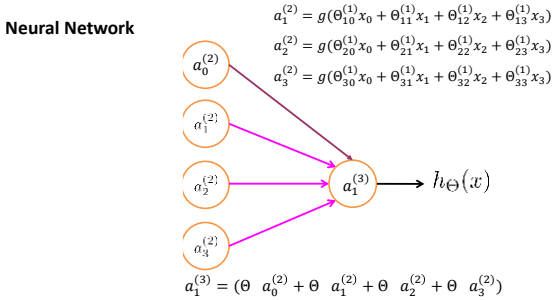
84


Neural Network
$a_1^{(2)} = g(\Theta_{10}^{(1)}x_0 + \Theta_{11}^{(1)}x_1 + \Theta_{12}^{(1)}x_2 + \Theta_{13}^{(1)}x_3)$
$a_2^{(2)} = g(\Theta_{20}^{(1)}x_0 + \Theta_{21}^{(1)}x_1 + \Theta_{22}^{(1)}x_2 + \Theta_{23}^{(1)}x_3)$
$a_3^{(2)} = g(\Theta_{30}^{(1)}x_0 + \Theta_{31}^{(1)}x_1 + \Theta_{32}^{(1)}x_2 + \Theta_{33}^{(1)}x_3)$
x_0
x_1
x_2
x_3
$a_0^{(2)}$
$a_1^{(2)}$
$a_2^{(2)}$
$a_3^{(2)}$
$a_1^{(3)}$
$h_\Theta(x)$

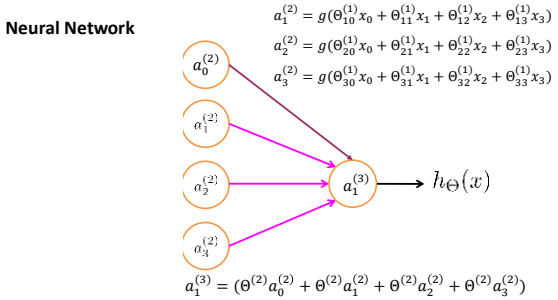
85


Neural Network
$a_1^{(2)} = g(\Theta_{10}^{(1)}x_0 + \Theta_{11}^{(1)}x_1 + \Theta_{12}^{(1)}x_2 + \Theta_{13}^{(1)}x_3)$
$a_2^{(2)} = g(\Theta_{20}^{(1)}x_0 + \Theta_{21}^{(1)}x_1 + \Theta_{22}^{(1)}x_2 + \Theta_{23}^{(1)}x_3)$
$a_3^{(2)} = g(\Theta_{30}^{(1)}x_0 + \Theta_{31}^{(1)}x_1 + \Theta_{32}^{(1)}x_2 + \Theta_{33}^{(1)}x_3)$
$a_0^{(2)}$
$a_1^{(2)}$
$a_2^{(2)}$
$a_3^{(2)}$
$a_1^{(3)}$
$h_\Theta(x)$
$a_1^{(3)} = ???$

86

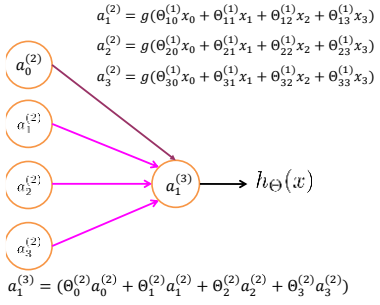

Neural Network
$a_1^{(2)} = g(\Theta_{10}^{(1)}x_0 + \Theta_{11}^{(1)}x_1 + \Theta_{12}^{(1)}x_2 + \Theta_{13}^{(1)}x_3)$
$a_2^{(2)} = g(\Theta_{20}^{(1)}x_0 + \Theta_{21}^{(1)}x_1 + \Theta_{22}^{(1)}x_2 + \Theta_{23}^{(1)}x_3)$
$a_3^{(2)} = g(\Theta_{30}^{(1)}x_0 + \Theta_{31}^{(1)}x_1 + \Theta_{32}^{(1)}x_2 + \Theta_{33}^{(1)}x_3)$
$a_0^{(2)}$
$a_1^{(2)}$
$a_2^{(2)}$
$a_3^{(2)}$
$a_1^{(3)}$
$h_\Theta(x)$
$a_1^{(3)} = (\Theta\ \ a_0^{(2)} + \Theta\ \ a_1^{(2)} + \Theta\ \ a_2^{(2)} + \Theta\ \ a_3^{(2)})$

87


Neural Network
$a_1^{(2)} = g(\Theta_{10}^{(1)}x_0 + \Theta_{11}^{(1)}x_1 + \Theta_{12}^{(1)}x_2 + \Theta_{13}^{(1)}x_3)$
$a_2^{(2)} = g(\Theta_{20}^{(1)}x_0 + \Theta_{21}^{(1)}x_1 + \Theta_{22}^{(1)}x_2 + \Theta_{23}^{(1)}x_3)$
$a_3^{(2)} = g(\Theta_{30}^{(1)}x_0 + \Theta_{31}^{(1)}x_1 + \Theta_{32}^{(1)}x_2 + \Theta_{33}^{(1)}x_3)$
$a_0^{(2)}$
$a_1^{(2)}$
$a_2^{(2)}$
$a_3^{(2)}$
$a_1^{(3)}$
$h_\Theta(x)$
$a_1^{(3)} = (\Theta^{(2)}a_0^{(2)} + \Theta^{(2)}a_1^{(2)} + \Theta^{(2)}a_2^{(2)} + \Theta^{(2)}a_3^{(2)})$

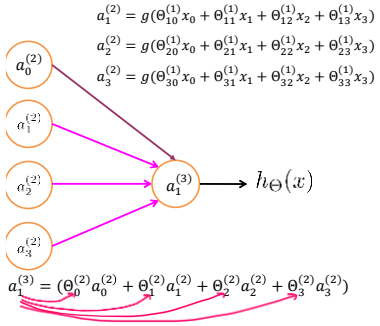
88

Neural Network



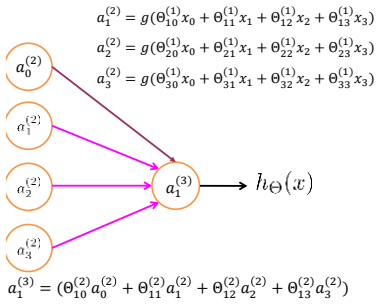
89

Neural Network



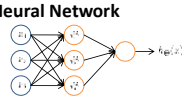
90

Neural Network



91

Neural Network



$a_i^{(j)}$ = "activation" of unit i in layer j

$\Theta^{(j)}$ = matrix of weights controlling function mapping from layer j to layer $j+1$

$$a_1^{(2)} = g(\Theta_{10}^{(1)}x_0 + \Theta_{11}^{(1)}x_1 + \Theta_{12}^{(1)}x_2 + \Theta_{13}^{(1)}x_3)$$
$$a_2^{(2)} = g(\Theta_{20}^{(1)}x_0 + \Theta_{21}^{(1)}x_1 + \Theta_{22}^{(1)}x_2 + \Theta_{23}^{(1)}x_3)$$
$$a_3^{(2)} = g(\Theta_{30}^{(1)}x_0 + \Theta_{31}^{(1)}x_1 + \Theta_{32}^{(1)}x_2 + \Theta_{33}^{(1)}x_3)$$
$$h_\Theta(x) = a_1^{(3)} = g(\Theta_{10}^{(2)}a_0^{(2)} + \Theta_{11}^{(2)}a_1^{(2)} + \Theta_{12}^{(2)}a_2^{(2)} + \Theta_{13}^{(2)}a_3^{(2)})$$

92

Neural Network



$a_i^{(j)}$ = "activation" of unit i in layer j

$\Theta^{(j)}$ = matrix of weights controlling function mapping from layer j to layer $j+1$

$$a_1^{(2)} = g(\Theta_{10}^{(1)}x_0 + \Theta_{11}^{(1)}x_1 + \Theta_{12}^{(1)}x_2 + \Theta_{13}^{(1)}x_3)$$
$$a_2^{(2)} = g(\Theta_{20}^{(1)}x_0 + \Theta_{21}^{(1)}x_1 + \Theta_{22}^{(1)}x_2 + \Theta_{23}^{(1)}x_3)$$
$$a_3^{(2)} = g(\Theta_{30}^{(1)}x_0 + \Theta_{31}^{(1)}x_1 + \Theta_{32}^{(1)}x_2 + \Theta_{33}^{(1)}x_3)$$
$$h_\Theta(x) = a_1^{(3)} = g(\Theta_{10}^{(2)}a_0^{(2)} + \Theta_{11}^{(2)}a_1^{(2)} + \Theta_{12}^{(2)}a_2^{(2)} + \Theta_{13}^{(2)}a_3^{(2)})$$

If network has s_j units in layer j, s_{j+1} units in layer $j+1$, then $\Theta^{(j)}$ will be of dimension $s_{j+1} \times (s_j + 1)$.

Quiz – 4

